XRPL Validator — Daily Engineering Report · 2026-10-20

Period: 2026-10-20 00:00 to 24:00 US Eastern (2026-10-20 04:00 to 2026-10-21 04:00 UTC); ledgers #107,775,996 to #107,797,696. Every count from the ledger and the validator's logs stops at #107,797,696, closed 03:59:51 UTC. Generated 2026-10-21 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded (all causes resolved or fixed) — coverage outside planned maintenance 99.898% (objective 99.9%): 22 ledgers unsigned.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,517 of the 21,701 ledgers the network closed (99.152%); every validation matched the network's validated ledger. Unsigned: 162 during planned restarts, 22 outside maintenance (3 incidents).
- **Native engine:** 2 engine receipts: #107,787,681 — fixed: F422, commit ca1f172, deployed in cycle 186 (2026-10-21 02:04 UTC); #107,794,217 — fixed: F423, commit 53c58cf, deployed in cycle 186 (2026-10-21 02:04 UTC).
- **Changes:** 3 deployments (cycles 184, 185, 186), 7 commits merged into main, 2 engine findings fixed (F422–F423).
- **Network:** 7 slow-close periods; the longest, 15:49 UTC–15:52 UTC, closed ledgers up to 50 s apart with 3 empty ledgers.
- **Independence program:** 0 of 30 qualifying days (snapshot at 04:20:37 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,701
Validations issued	21,517
Coverage (validations / ledgers closed)	99.152%
Validations matching the network's validated ledger hash	21,517 of 21,517
Unsigned during planned restarts	162
Unsigned outside maintenance	22

Measure	Value
Coverage outside planned maintenance (objective 99.9%)	99.898%
Restarts	3
Basis: the hash the trusted quorum named	11,674
Basis: our own ledger hash	9,680
Basis: our own hash, witnessed by the reference node (quorum short)	163
External agreement, data.xrpl.org (UTC day 2026-10-20)	0.99430 (104 missed of 18,239)

Restarts:

- 13:36:47 UTC: deployment of cycle 184; validations resumed at #107,784,820 (9 ledgers unsigned while the validator warmed up)
- 16:59:18 UTC: deployment of cycle 185; validations resumed at #107,787,817 (69 ledgers unsigned while the validator warmed up)
- 02:03:21 UTC: deployment of cycle 186; validations resumed at #107,795,984 (84 ledgers unsigned while the validator warmed up)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,775,996–#107,797,696), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	2
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	156
Step 5 · own hash	2
Writer	2

Live counters, snapshot at 04:20:37 UTC (since the last restart, 2026-10-21 02:03 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	2,023 ledgers, 292,162 transactions	2,023 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	2,027 ledgers, 297,514 records	297,514 exact; differ 0, tree 0
3 · transaction order	2,027 ledgers (mode: on)	2,027 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	2,080 ledgers fetched	2,080 matched; differ 0, failed 0
4 · trusted validations	2,109 ledgers judged	2,094 agreed; differ 0, short of quorum 15
5 · our own ledger hash	2,111 ledgers	2,027 own; differ 0, partial 84
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	12:30 UTC	#107,783,807–#107,783,814 (8)	Step 4 · validations	26–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (none signed); see INC-01
R-02	15:07 UTC	#107,786,161–#107,786,196 (36)	Step 4 · validations	25–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-03	15:15 UTC	#107,786,260–#107,786,262 (3)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-04	15:49 UTC	#107,786,761–#107,786,772 (12)	Step 4 · validations	0–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (none signed); see INC-02
R-05	16:54 UTC	#107,787,681	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	AccountDelete tx DE319724D91B: DirectoryNode 27BA7B2B (deleted on the network): absent from our record	fixed: F422, commit ca1f172, deployed in cycle 186 (2026-10-21 02:04 UTC)
R-06	17:18 UTC	#107,788,027–#107,788,030 (4)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-07	18:01 UTC	#107,788,680–#107,788,682 (3)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-08	19:10 UTC	#107,789,725–#107,789,734 (10)	Step 4 · validations	19–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-09	19:12 UTC	#107,789,736–#107,789,740 (5)	Step 4 · validations	14–20 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-10	21:51 UTC	#107,792,109–#107,792,132 (24)	Step 4 · validations	26–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-11	23:00 UTC	#107,793,133–#107,793,151 (19)	Step 4 · validations	23–28 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-12	00:09 UTC	#107,794,163–#107,794,173 (11)	Step 4 · validations	26–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

ID	Time	Ledger(s)	Step	Description	Disposition
R-13	00:12 UTC	#107,794,217	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	OfferCancel tx A9E17FDC53F0: DirectoryNode 083659DD (modified on the network): FinalFields.IndexNext ours="0000000000000007" theirs="00000…	fixed: F423, commit 53c58cf, deployed in cycle 186 (2026-10-21 02:04 UTC); see INC-03
R-14	00:49 UTC	#107,794,787–#107,794,792 (6)	Step 4 · validations	17–23 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-15	02:32 UTC	#107,796,352–#107,796,366 (15)	Step 4 · validations	26–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-20-01, INC-01 is INC-2026-10-20-01.

Incidents

INC-01 · 8 ledgers unsigned

Window	12:30:20 UTC to 12:31:00 UTC
Ledgers	#107,783,807–#107,783,814 (8)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-01
Cause	Since cycle 183 the validator signs a ledger only once the trusted validations reaching it reach quorum. On mainnet #107783807-814 (20 Oct) only 26 or 27 of quorum 28 reached us, so it signed none of the eight, though the reference node validated each and data.xrpl.org scores them missed.
Action	commit e6bb9d1 — fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees; deployed in cycle 184 at 2026-10-20 13:37 UTC

INC-02 · 12 ledgers unsigned

Window	15:49:10 UTC to 15:52:21 UTC
Ledgers	#107,786,761–#107,786,772 (12)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-04
Cause	Network-wide slow closes (15:49 UTC–15:52 UTC: up to 50 s apart, 3 empty ledgers) during a burst of Batch transactions (#107,786,760: 1,122 of 1,225 transactions; #107,786,762: 1,364 of 1,536 transactions). On mainnet #107786761-772 (20 Oct) three heavy ledgers (1,001 to 1,536 transactions) and a burst of peer reconnects left the validator's ledger feed 11 ledgers behind and our tally of trusted validations short on those ledgers. The quorum on #107786773 came first, the validator signed it, and a validator cannot sign an older ledger after a newer one, so twelve went unsigned, though our own hash and the reference node's agreed on each a few seconds later.
Action	commit 00cce16 — fix(validator): while the validator's ledger feed catches up, sign the older ledgers in order, not jump to a newer quorum; deployed in cycle 185 at 2026-10-20 17:00 UTC

INC-03 · 2 ledgers unsigned

Window	00:12:32 UTC to 00:12:40 UTC
Ledgers	#107,794,217–#107,794,218 (2)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-13
Cause	rippled's dirRemove has two branches for an empty last page that older rippled code left behind. When a page in the middle of a directory empties and the page after it is the last one and empty, that page is erased too: the previous page becomes the last, and the root points back to it. When the root's last entry leaves and both its links name one empty page, the root's links drop to 0 and that page is erased, then the root unless kept. The engine unlinked only the emptied page.
Action	commit 53c58cf — fix(ledger): F423 — removing a directory page takes an empty last page after it, as rippled's dirRemove does (ApplyView.cpp:362); deployed in cycle 186 at 2026-10-21 02:04 UTC

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
184	`e6bb9d1`	13:37:37 UTC	2,163 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	391 matched, 0 mismatched	#59 from 14:04 UTC: ended early: superseded by the next deployment (2026-10-20 17:00 UTC)
185	`00cce16`	17:00:09 UTC	2,171 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	385 matched, 0 mismatched	#60 from 17:26 UTC: ended early: superseded by the next deployment (2026-10-21 02:04 UTC)
186	`53c58cf`	02:04:14 UTC	2,181 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	407 matched, 0 mismatched	#61 from 02:30 UTC: in progress

Cycle 184 delivered: `e6bb9d1` fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees

Cycle 185 delivered: `7866189` feat(validator): independence 5a, sign a ledger only with our own hash of it; `00cce16` fix(validator): while the validator's ledger feed catches up, sign the older ledgers in order, not jump to a newer quorum

Cycle 186 delivered: `77e4bca` feat(validator): Step 4 stage A, make the engine's inputs from a peer ledger's bytes and hold them against the reference node's; `3c4e4fa` feat(validator): a short ledger's receipt names the trusted validators whose validations never reached us; `ca1f172` fix(ledger): F422 — AccountDelete takes an old account's empty last directory page with its directory (ApplyView.cpp:221); `53c58cf` fix(ledger): F423 — removing a directory page takes an empty last page after it, as rippled's dirRemove does (ApplyView.cpp:362)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
12:57 UTC	xrpl-validator	`e6bb9d1`	fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees
14:12 UTC	xrpl-validator	`7866189`	feat(validator): independence 5a, sign a ledger only with our own hash of it

Merged	Repo	Commit	Subject
16:19 UTC	xrpl-validator	`00cce16`	fix(validator): while the validator's ledger feed catches up, sign the older ledgers in order, not jump to a newer quorum
01:12 UTC	xrpl-validator	`3c4e4fa`	feat(validator): a short ledger's receipt names the trusted validators whose validations never reached us
01:12 UTC	xrpl-validator	`77e4bca`	feat(validator): Step 4 stage A, make the engine's inputs from a peer ledger's bytes and hold them against the reference node's
01:12 UTC	xrpl-validator	`ca1f172`	fix(ledger): F422 — AccountDelete takes an old account's empty last directory page with its directory (ApplyView.cpp:221)
01:12 UTC	xrpl-validator	`53c58cf`	fix(ledger): F423 — removing a directory page takes an empty last page after it, as rippled's dirRemove does (ApplyView.cpp:362)

Findings fixed

- **fix(ledger): F422 — AccountDelete takes an old account's empty last directory page with its directory (ApplyView.cpp:221)** (`ca1f172` , xrpl-validator) rippled removes a deleted account's owner directory through ApplyView::emptyDirDelete, which also clears a case older rippled code left behind: an empty last page. When the root holds no entries and both its links name one other page that is empty too, the root's links drop to 0 and that page is erased, then the root. The engine deleted the empty root only.
- **fix(ledger): F423 — removing a directory page takes an empty last page after it, as rippled's dirRemove does (ApplyView.cpp:362)** (`53c58cf` , xrpl-validator) rippled's dirRemove has two branches for an empty last page that older rippled code left behind. When a page in the middle of a directory empties and the page after it is the last one and empty, that page is erased too: the previous page becomes the last, and the root points back to it. When the root's last entry leaves and both its links name one empty page, the root's links drop to 0 and that page is erased, then …

Other engineering changes

- **fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees** (`e6bb9d1` , xrpl-validator) Since cycle 183 the validator signs a ledger only once the trusted validations reaching it reach quorum. On mainnet #107783807-814 (20 Oct) only 26 or 27 of quorum 28 reached us, so it signed none of the eight, though the reference node validated each and data.xrpl.org scores them missed.
- **feat(validator): independence 5a, sign a ledger only with our own hash of it** (`7866189` , xrpl-validator) Until now the validator signed the hash the trusted validators reached quorum on (cycle 183), or the reference node's when that quorum fell short here and our own hash agreed (cycle 184). Our own hash of the ledger was checked only in that second case.
- **fix(validator): while the validator's ledger feed catches up, sign the older ledgers in order, not jump to a newer quorum** (`00cce16` , xrpl-validator) On mainnet #107786761-772 (20 Oct) three heavy ledgers (1,001 to 1,536 transactions) and a burst of peer reconnects left the validator's ledger feed 11 ledgers behind and our tally of trusted validations short on those ledgers. The quorum on #107786773 came first, the validator signed it, and a validator cannot sign an older ledger after a newer one, so twelve went unsigned, though our own hash and the reference node's agreed on each a few seconds …
- **feat(validator): a short ledger's receipt names the trusted validators whose validations never reached us** (`3c4e4fa` , xrpl-validator) When fewer than a quorum of trusted validations reach us for a ledger, the receipt now

lists the trusted validators (nH… keys) whose validation of that hash never arrived, those that validated another hash among them, and says whether the ledger was signed anyway.

- **feat(validator): Step 4 stage A, make the engine's inputs from a peer ledger's bytes and hold them against the reference node's** (`77e4bca`, xrpl-validator) Step 4 needs the engine to run on ledger data from peers, not the reference node's JSON. Stage A builds that copy and checks it live, in shadow only: the engine still runs on the reference node's inputs and peer_fetch::is_source() stays false.

Independence program

Qualifying days: **0 of 30** (snapshot at 04:20:37 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-20 — tracked #107,787,817–#107,794,028	Value
Validations	6,212 (6,212 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	no

Not qualifying: the ledger data came from the reference node, not peers (Step 4).

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,701 (#107,775,996–#107,797,696)
Mean close interval	3.98 s
Longest close interval	50 s (before #107,786,762)
Closes at least 15 s apart	17
Empty ledgers	3

- Slow closes 15:08:11 UTC–15:08:30 UTC: ledgers #107,786,166, 1 close at least 15 s apart, up to 19 s.
- Slow closes 15:49:10 UTC–15:52:10 UTC: ledgers #107,786,762–#107,786,770 (9), 5 closes at least 15 s apart, up to 50 s; empty: #107,786,769, #107,786,770, #107,786,771.
- Slow closes 19:10:40 UTC–19:13:10 UTC: ledgers #107,789,729–#107,789,740 (12), 6 closes at least 15 s apart, up to 20 s.
- Slow closes 23:45:02 UTC–23:45:20 UTC: ledgers #107,793,807, 1 close at least 15 s apart, up to 18 s.
- Slow closes 00:49:51 UTC–00:50:20 UTC: ledgers #107,794,788, 1 close at least 15 s apart, up to 29 s.
- Slow closes 00:50:41 UTC–00:51:40 UTC: ledgers #107,794,792–#107,794,794 (3), 2 closes at least 15 s apart, up to 39 s.
- Slow closes 00:51:43 UTC–00:52:00 UTC: ledgers #107,794,798, 1 close at least 15 s apart, up to 17 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Soak #61 (cycle 186) in progress since 2026-10-21 02:30 UTC; due to complete 2026-10-22 02:30 UTC.
- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

- **2026-10-19 report:** it did not record that the validator issued no validation for ledger #107,770,853 (22:20:10 UTC), because the format-v2 report had no service-level section. The trusted validations for #107,770,853 reached quorum only after those for #107,770,854. Under cycle 183 the validator signed only on a quorum, so it signed #107,770,854 first and could no longer sign the older ledger. Every other ledger that day was signed (22,147 of 22,148, 99.995%), and every validation matched the network. Fixed by commit e6bb9d1, deployed in cycle 184 on 2026-10-20 at 13:37 UTC. The 2026-10-19 report will be re-issued in format v3.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-20 12:54	10-20 12:57	xrpl-validator	`e6bb9d1`	fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees
10-20 14:08	10-20 14:12	xrpl-validator	`7866189`	feat(validator): independence 5a, sign a ledger only with our own hash of it
10-20 16:17	10-20 16:19	xrpl-validator	`00cce16`	fix(validator): while the validator's ledger feed catches up, sign the older ledgers in order, not jump to a newer quorum
10-20 18:57	10-21 01:12	xrpl-validator	`3c4e4fa`	feat(validator): a short ledger's receipt names the trusted validators whose validations never reached us
10-20 18:57	10-21 01:12	xrpl-validator	`77e4bca`	feat(validator): Step 4 stage A, make the engine's inputs from a peer ledger's bytes and hold them against the reference node's
10-20 21:42	10-21 01:12	xrpl-validator	`ca1f172`	fix(ledger): F422 — AccountDelete takes an old account's empty last directory page with its directory (ApplyView.cpp:221)
10-21 00:20	10-21 01:12	xrpl-validator	`53c58cf`	fix(ledger): F423 — removing a directory page takes an empty last page after it, as rippled's dirRemove does (ApplyView.cpp:362)

Times in UTC. A commit appears on the day it was committed to main or merged into it.